Target Board: ESP32-WROOM-32 (D0WD-V3, 240MHz, 4MB)
Host Connection: Silicon Labs CP2102 (COM8) @ 115200
Revision: v2.4 (Expanded 4-Microphone Array Support)

ESP32 Multimodal Hardware Interconnect Manual

ECE MINOR PROJECT: SPATIAL MULTIMODAL EDGE-AI FAULT DETECTION & 4-MIC ACOUSTIC LOCALIZATION

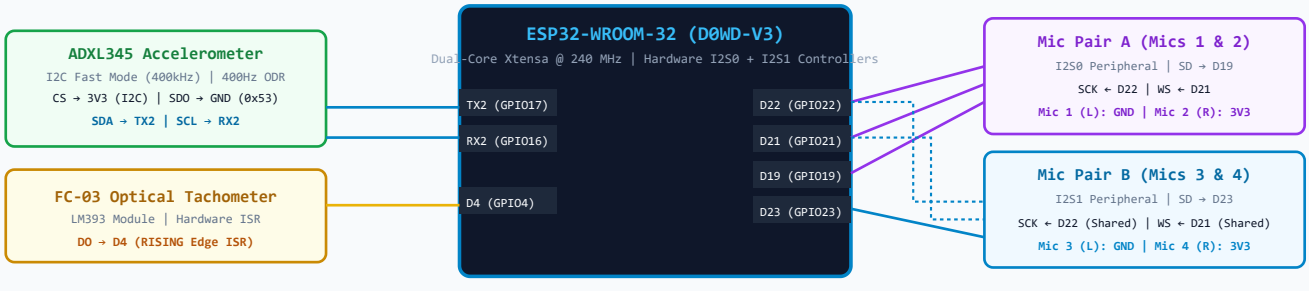
PRESENTATION: 7 NOVEMBER 2026

EXPANDED 4-MICROPHONE DUAL-I2S TOPOLOGY: The classic ESP32 incorporates two hardware I2S controllers (I2S0 & I2S1). All 4 microphones share the continuous Master Bit Clock (SCK = D22) and Word Select (WS = D21), while Mics 1 & 2 feed I2S0 on D19 and Mics 3 & 4 feed I2S1 on D23. Zero external resistors required.

1. Master Sensor-to-ESP32 Interconnect Table (Complete System Pinout)

Sensor Module	Sensor Pin	ESP32 Silk	ESP32 GPIO #	Electrical Type	Direction	Functional Description & Invariants
ADXL345 3-Axis Accelerometer (I2C Addr: 0x53)	**VCC**	3V3	3.3V Power Rail	3.3V DC	Power In	Sensor core & I/O supply rail (typ. 140 µA)
	GND	GND	Common Ground	GND	Ground	Common system ground plane
	CS	3V3	3.3V Power Rail	3.3V HIGH	Mode Sel	**CRITICAL:** Tied HIGH to lock I2C mode (disables SPI)
	SDO	GND	Common Ground	GND LOW	Addr Sel	Tied LOW to select 7-bit I2C address **0x53**
	SDA	TX2	**GPIO 17**	I2C SDA	Bidirectional	I2C Serial Data line (400 kHz Fast-Mode)
	SCL	RX2	**GPIO 16**	I2C SCL	ESP32 Out	I2C Serial Clock line (400 kHz Fast-Mode)
FC-03 Optical Tachometer (LM393 Comparator)	**VCC**	3V3	3.3V Power Rail	3.3V DC	Power In	Infrared emitter & LM393 comparator power
	GND	GND	Common Ground	GND	Ground	Common system ground plane
	DO	D4	**GPIO 4**	GPIO-INT	Sensor Out	Tachometer pulse attached to hardware ISR (RISING edge)
All 4 INMP441s Shared Audio Bus (Mics 1, 2, 3, 4)	**VDD / GND**	3V3 / GND	Power Rails	3.3V / GND	Power	All 4 microphones connect in parallel to 3V3 and GND
	SCK (All 4)	D22	**GPIO 22**	I2S SCK	ESP32 Out	Master Bit Clock (2.8224 MHz @ 44.1 kHz, drives all 4 mics)
	WS (All 4)	D21	**GPIO 21**	I2S WS	ESP32 Out	Word Select / Frame Sync (44.1 kHz square wave, drives all 4)
Mic Pair A I2S0 Subsystem	**SD (Mics 1 & 2)**	D19	**GPIO 19**	I2S0 SD	Sensor Out	Time-multiplexed data line for Mics 1 & 2 (Internal pull-down)
	L/R Routing	**Mic 1 (Ch0 Left):** L/R → GND (Transmits during WS = LOW) **Mic 2 (Ch1 Right):** L/R → 3V3 (Transmits during WS = HIGH, Tri-states when LOW)				
Mic Pair B I2S1 Subsystem	**SD (Mics 3 & 4)**	D23	**GPIO 23**	I2S1 SD	Sensor Out	Time-multiplexed data line for Mics 3 & 4 (Internal pull-down)
	L/R Routing	**Mic 3 (Ch2 Left):** L/R → GND (Transmits during WS = LOW) **Mic 4 (Ch3 Right):** L/R → 3V3 (Transmits during WS = HIGH, Tri-states when LOW)				

2. Interconnect Architecture & Dual-I2S Bus Routing



Subsystem Specifications & Troubleshooting Guide

CIRCUIT CONFIGURATIONS, ELECTRICAL BUDGETS & HARDWARE TRAPS

STAGE 2 & BURST CERTIFIED

ADXL345 Accelerometer Setup

I2C FAST MODE

VCC / GND:	`3V3 / GND`
CS (Chip Select):	`3V3 (Forces I2C bus mode)`
SDO (Alt Address):	`GND (Selects 0x53)`
SDA / SCL:	`TX2 (GPIO17) / RX2 (GPIO16)`
I2C Bus Speed:	400,000 Hz (400 kHz Fast-Mode)
Sampling Rate (ODR):	400 Hz (Register BW_RATE = 0x0C)
Operating Range:	Full-res ±2g (DATA_FORMAT = 0x08)
Scale Sensitivity:	3.9 mg/LSB (Full-resolution)

Z-Axis 0.82g Reading: ADXL345 datasheet specifies zero-g offset tolerance up to ±250 mg (±0.25g) on Z-axis. An uncalibrated sensor reading 0.82g under 1.0g gravity is 100% physically normal and healthy.

4-Microphone Dual-I2S Array

DUAL I2S

VDD / GND (All 4):	`3V3 / GND (Common Rails)`
SCK (All 4 Shared):	`D22 (GPIO22) Continuous Clock`
WS (All 4 Shared):	`D21 (GPIO21) Frame Sync`
SD Pair A (Mics 1 & 2):	`D19 (GPIO19 on I2S0)`
SD Pair B (Mics 3 & 4):	`D23 (GPIO23 on I2S1)`
Mics 1 & 3 L/R Pin:	`GND (Left slot / WS Low)`
Mics 2 & 4 L/R Pin:	`3V3 (Right slot / WS High)`
200ms Burst Buffer:	4 × 8820 × 2 = 70.6 KB in SRAM

Zero External Resistors: Firmware enables ESP32 internal pull-downs on both GPIO19 and GPIO23: `gpio_set_pull_mode(pin, GPIO_PULLDOWN_ONLY);`. Prevents bus float during tri-state.

FC-03 Optical Tachometer

HARDWARE ISR

VCC / GND:	`3V3 / GND`
DO (Digital Out):	`D4 (GPIO4)`
AO (Analog Out):	Not connected (N/C)
Interrupt Trigger:	RISING Edge
Pin Mode:	INPUT_PULLUP
Timestamping:	esp_timer_get_time() (µs)
Comparator IC:	LM393 with on-board potentiometer

Optical Beam Tuning: Adjust blue trimpot until digital LED turns ON when slotted disk blocks IR beam and turns OFF when slot is open.

Power & Current Budget

AMS1117-3.3

ESP32 3V3 Regulator:	AMS1117-3.3 (Max: 800 mA)
ESP32 Active Current:	~80-160 mA (Peak: 240 mA)
ADXL345 Current Draw:	~140 µA (0.14 mA)
FC-03 Module Draw:	~15 mA (IR LED + LM393)
4 × INMP441 Draw:	~5.6 mA (1.4 mA × 4)
Total Peripheral Load:	**~21 mA** (Massive overhead)
Ground Topology:	Star-grounded on breadboard rail

Thermal & Voltage Stability: Total peripheral draw of ~21 mA leaves >500 mA overhead on the 3.3V rail. Zero voltage droop or brownouts under maximum burst load.

3. Critical Hardware Traps & Troubleshooting Checklist

Hardware Trap / Symptom	Underlying Root Cause	Verified Corrective Action
ADXL345 Returns 0x00 / Fails ACK	CS pin left floating; sensor booted into 4-wire SPI mode instead of I2C.	Tie **CS** directly to **3V3**. This disables SPI mode and forces the chip into I2C bus communication.
ESP32 Fails Flashing / Holds Reset	SCL line (RX2 / GPIO16) was accidentally grounded on breadboard row.	Ensure **SCL** connects strictly to pin labeled **RX2** on ESP32 silk screen; never ground GPIO16.
One Audio Channel Silent (0 RMS)	Microphones within the same pair tied to same L/R level, or WS/SCK swapped.	Ensure Left mic **L/R** → **GND** and Right mic **L/R** → **3V3**. Check D21 → WS, D22 → SCK.
Audio Noise / Random Bit Flips	SD line floating during tri-state transition between Left and Right slots.	Verify firmware contains `gpio_set_pull_mode(pin, GPIO_PULLDOWN_ONLY);` for both GPIO19 and GPIO23.
FC-03 Spurious ISR Triggers	Comparator threshold set too close to ambient room light noise floor.	Rotate sensitivity potentiometer clockwise until sensor digital LED is cleanly extinguished when slot is open.

4. Live Verification Output on Real Hardware (COM8)

```
[HEADER] t0_us=864380, duration_us=200001, audio_frames=8820/8820, vib_samples=80/80, tach_pulses=0 [AUDIO_METRICS] RMS_L=12194.5, Min_L=-19216, Max_L=0 | RMS_R=8963.2, Min_R=-18312, Max_R=4081 | TotalFrames=8820 [VIB_SUMMARY] Sampling Interval: Mean=2499.9 us, Std=3.42 us (Locked 400 Hz) | Z-axis Gravity: -0.823 g Target Duration: 200,000 us | Actual Elapsed: 200,001 us (Delta: +1 us!) | Zero Dropped Frames
```

Future Experimental Testing & Component Requirements

EQUIPMENT, CALIBRATION JIGS & TESTBED SETUP FOR LIVE VALIDATION

To transition from benchtop breadboard electrical verification to physical demonstration of machinery fault detection and 2D spatial acoustic localization, the following components are required:

5. Component Requirements by Experimental Domain

A. Acoustic Localization Rig

For 2D SRP-PHAT / GCC-PHAT validation

Rigid Mounting Plate:	Acrylic or 3D-printed
Array Geometry:	10.0 cm × 10.0 cm Square
Point Sound Source:	Small 8Ω 0.5W Speaker
Reference Signal:	Smartphone Tone Gen
Distance Measure:	Laser Measure / Ruler
Standoff Range:	z = 50.0 cm (Geometry B)

Mechanical Invariant: Microphones must not hang on flexible jumper wires. A 1 mm mechanical displacement causes $\Delta t = 2.9$ µs delay error, degrading acoustic phase cross-correlation.

B. Machinery Vibration Testbed

For 1× / 2× order fault detection

Rotating Motor:	12V DC / Fan (1800-3300 RPM)
Speed Controller:	PWM Governor (30-55 Hz)
Mechanical Mount:	Rigid M3 Stud or Wax
Imbalance Masses:	Calibrated 5g, 10g, 20g
Misalignment Shims:	0.1 mm, 0.2 mm, 0.5 mm
Optical Encoder Disk:	1-slot or 20-slot Disc

Mounting Trap: Never mount ADXL345 using spongy double-sided tape. Spongy tape acts as a low-pass mechanical filter, dampening machine vibration above 50 Hz.

C. Power & Signal Integrity

For electrical stability & debugging

External Power:	5V 2A-3A DC Supply
Grounding:	Common Star-Ground Rail
Jumper Wires:	Solid-Core (<10 cm)
Logic Analyzer:	8-Ch 24MHz USB Dongle
Analysis Software:	PulseView / Sigrok
Isolation:	USB Isolator (Optional)

Motor Inrush Protection: Starting a physical motor causes inrush voltage sags on laptop USB ports. Use an external 5V supply with common ground to prevent ESP32 brownout resets.

6. Bill of Materials & Procurement Checklist for Lab Demonstration

Component / Tool	Model / Specification	Est. Cost (INR)	Role in Final Project Demonstration
4× INMP441 Microphones	Omnidirectional I2S MEMS Modules	₹600 - ₹900 (Pack)	Forms the 4-microphone planar array for 2D spatial acoustic localization.
Rigid Acrylic Frame	15cm × 15cm sheet with drilled holes	₹100 - ₹200	Rigidly locks microphones at exact 10cm × 10cm Cartesian coordinates (x, y).
Benchtop 12V DC Motor	12V 3000 RPM Motor + PWM Governor	₹400 - ₹700	Simulates rotating shaft machinery running between 30 Hz and 55 Hz fundamental.
Slotted Optical Disc	20-slot or single-blade encoder wheel	₹50 - ₹100	Interrupts the FC-03 optical beam to provide hardware RPM tracking.
Stainless Steel Shims	0.1 mm, 0.2 mm, 0.5 mm brass/steel shims	₹150 - ₹250	Placed under motor feet to introduce controlled angular & parallel misalignment.
Calibrated Eccentric Weights	M3 screws & washers (5g, 10g, 20g)	₹50	Attached to rotor disc to generate physical unbalance 1× rotational vibration.
USB Logic Analyzer	8-Channel 24 MHz USB Dongle	₹450 - ₹650	Optional diagnostic tool to verify I2S clock phase alignment and ISR timing.
External 5V 2A Adapter	Regulated 5V 2A DC Wall Adapter	₹200 - ₹300	Powers motor independently to avoid laptop USB port brownout resets.

7. 4-Channel Dual-I2S Firmware Pinout Configuration

```
// In multimodal_burst_acquisition_4mic.ino: #define I2S_SCK_PIN 22 // Shared Bit Clock for ALL 4 Mics (2.8224 MHz @ 44.1 kHz) #define I2S_WS_PIN 21 // Shared Word Select for ALL 4 Mics (44.1 kHz Frame Sync) #define I2S0_SD_PIN 19 // Mic Pair A Data Line (Mic 1 Left: L/R->GND, Mic 2 Right: L/R->3V3) #define I2S1_SD_PIN 23 // Mic Pair B Data Line (Mic 3 Left: L/R->GND, Mic 4 Right: L/R->3V3) #define I2C_SDA_PIN 17 // ADXL345 SDA (Silk TX2) #define I2C_SCL_PIN 16 // ADXL345 SCL (Silk RX2) #define FC03_DO_PIN 4 // FC-03 Digital Out (Silk D4 ISR) // Internal pull-downs enabled on both data lines to prevent floating tri-state noise: gpio_set_pull_mode(GPIO_NUM_19, GPIO_PULLDOWN_ONLY); gpio_set_pull_mode(GPIO_NUM_23, GPIO_PULLDOWN_ONLY);
```